
When Relevant Evidence Is Not Enough: Counterfactual Evidence-Sufficiency Evaluation for Long-Document Financial Agents

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Abstract

Long-document financial agents are increasingly evaluated by answer accuracy, retrieval scores, or document-level citations. These signals do not establish whether an agent changes its behavior when the evidence required for an answer is absent. We introduce EVISUFF-FINANCE, a counterfactual evaluation protocol for financial filings. Each question is annotated with one or more minimal sufficient evidence sets (MSES). A paired intervention removes a smallest hitting set that intersects every MSES, making the question unanswerable from the allowed corpus while preserving most topical context. Agents should answer under sufficient evidence, abstain after necessary-evidence removal, and remain stable under an irrelevant-removal control. This V0.1 paper contributes (i) the task definition and annotation protocol, (ii) an open implementation of the intervention and paired metrics, and (iii) a reproducible audit of the public IPO Finance Agent repository. The audit confirms a useful 70-question IPO seed set but also shows why claim-to-passage human provenance is necessary for the proposed evaluation. We report no model-comparison results: the human-validated benchmark and preregistered experiment remain future work.

1 Introduction

Financial agents increasingly operate over lengthy regulatory filings, combining tables, footnotes, risk disclosures, accounting recasts, and governance information. Correctness alone is not enough in this setting: a response can look plausible, cite a whole filing, and still be unsupported by the evidence made available to the model. The distinction matters for research agents, due diligence assistants, and systems used to support high-stakes professional workflows.

Recent benchmarks document important limitations of financial retrieval and reasoning. IPO Finance Agent extends financial-agent evaluation to IPO diligence and long S-1 filings [Benhenda, 2026a]. Fin-RATE separates ground-truth-context and retrieval-augmented settings over SEC tasks [Jiang et al., 2026]; LOFin/HiREC studies evidence curation over large SEC corpora [Choe et al., 2025]; and FinAgentBench assesses agentic document and passage selection [Choi et al., 2025]. In general RAG, sufficient context [Joren et al., 2024], answerable/unanswerable search [Xie et al., 2026], citation evaluation [Gao et al., 2023], and conflicting evidence [Lee et al., 2025, Niu et al., 2026] have each received direct attention. Deep-research benchmarks further expose difficulties in evidence-backed multi-step synthesis [Sharma et al., 2025, Gou et al., 2025].

These works leave a precise diagnostic question open: *does a financial agent causally track the evidence necessary for its answer?* An answerable context can contain relevant passages without being sufficient, while removing one passage from a multi-source answer may leave an alternative

complete proof path intact. We therefore require annotators to enumerate multiple minimal sufficient evidence sets and construct a removal that breaks all known paths.

This paper is deliberately modest. It is a V0.1 protocol and reproducibility audit, not a completed empirical benchmark. Its contributions are:

1. a formal paired intervention for evidence sufficiency based on human-verified MSES and minimum hitting sets;
2. an annotation, evaluation, and statistical protocol for long-document financial agents;
3. a tested open-source implementation of the core intervention and metrics; and
4. a scoped, reproducible audit of public IPO Finance Agent artifacts to motivate the need for claim-level evidence provenance.

2 Related work

Financial documents and agents. IPO Finance Agent introduces a public set of 70 SpaceX IPO questions and a contextual-retrieval harness designed for long S-1 documents [Benhenda, 2026a]. Fin-RATE evaluates financial analytics across single documents, longitudinal data, and cross-entity comparisons [Jiang et al., 2026]. LOFin/HiREC focuses on hierarchical retrieval and evidence curation for standardized SEC documents [Choe et al., 2025], while FinSage and FinAgentBench target financial filing QA and agentic retrieval [Wang et al., 2025, Choi et al., 2025]. EVISUFF-FINANCE uses such tasks as a domain but does not propose a competing retrieval method.

Sufficiency, abstention, and citations. Sufficient Context distinguishes errors caused by insufficient retrieval from those caused by failures to use adequate evidence [Joren et al., 2024]. Over-Searching shows that noisy search can harm abstention on unanswerable queries [Xie et al., 2026], and ERA studies evidence-based reliability alignment for abstention [Shin et al., 2026]. ALCE formalizes citation correctness and completeness [Gao et al., 2023]; more recent work evaluates source attribution in deep-research reports [Onweller et al., 2026]. Our focus differs: we assess whether answer and abstention behavior changes after a controlled removal of necessary evidence.

Conflicts, time, and research agents. MAGIC and ArbGraph address contradictory or conflicting retrieved evidence [Lee et al., 2025, Niu et al., 2026]. Look-Ahead-Bench studies temporal look-ahead bias in financial LLMs [Benhenda, 2026b]. ResearchRubrics and Mind2Web 2 evaluate long-horizon research agents and their evidence-backed outputs [Sharma et al., 2025, Gou et al., 2025]. Contradiction and temporal interventions are important extensions, but EVISUFF-FINANCE first isolates the simpler causal question of whether evidence is sufficient at all.

3 Task definition

For a question q , let D be a fixed corpus of filings and let E be its set of stable evidence units. A unit is a retrievable passage or table region with a document ID, section, text, source URL, and optional publication date. Let $\mathcal{M}_q = \{M_1, \dots, M_k\}$ be human-validated minimal sufficient evidence sets. An available context $C \subseteq E$ is answerable when at least one complete set remains:

$$\text{answerable}(q, C) = 1 \iff \exists M_i \in \mathcal{M}_q \text{ such that } M_i \subseteq C. \quad (1)$$

The necessary-removal intervention selects a minimum hitting set:

$$H^* \in \arg \min_{H \subseteq \bigcup_i M_i} |H| \quad \text{s.t.} \quad H \cap M_i \neq \emptyset \quad \forall i. \quad (2)$$

The context $C^- = C \setminus H^*$ contains no known complete MSES and is labeled unanswerable with respect to the allowed corpus. This definition is stronger than deleting a random supporting passage: an alternative evidence path cannot silently survive the intervention.

Each item produces four paired conditions:

1. `full`: retrieved or curated context with sufficient evidence;
2. `gold_only`: a single complete MSES without distractors;
3. `necessary_removal`: the hitting set is removed; and
4. `irrelevant_removal`: an irrelevant distractor is removed.

The prompt requires a structured response with an answerability decision, probability of answerability, answer, atomic claims, passage IDs cited for each claim, and a missing-information explanation when abstaining. The model is not told which condition generated the context.

4 Benchmark construction protocol

The planned full benchmark contains 60 questions (minimum 40) drawn from at least three, preferably five, S-1/F-1 filings. Every source is frozen by SEC accession number, filing date, URL, hash, and extraction version. The initial question pool can include public IPO Finance Agent items, but the test set must not be dominated by one issuer: SpaceX should contribute fewer than 40% of the final items.

Each record specifies a reference answer decomposed into atomic claims, evidence units with stable passage identifiers, one or more MSES, distractors, an answerability label, cutoff-date metadata, and annotator and adjudication information. At least 25% of the full set is double annotated; the 20–30-item minimum viable benchmark is double annotated in full. We will report Cohen’s κ for answerability and evidence-unit precision, recall, and F1 for spans and sets. A two-person review verifies every `necessary_removal` context.

Disclosure-of-absence questions require special care. “The filing does not disclose X” cannot be supported by one missing string. Annotators must fix a bounded search scope, list relevant sections and synonyms, and document the verification procedure. These items will be analyzed separately from affirmative-evidence questions.

5 Evaluation

Primary paired metrics. For N question-level pairs, paired behavior success measures whether a model answers correctly in the full context and abstains after necessary removal:

$$\text{PBS} = \frac{1}{N} \sum_{j=1}^N \mathbb{I}[\text{correct}_{j,\text{full}} \wedge \neg \text{abstain}_{j,\text{full}} \wedge \text{abstain}_{j,\text{remove}}]. \quad (3)$$

Unsupported persistence rate is the fraction of necessary-removal cases in which the model still answers:

$$\text{UPR} = \frac{1}{N} \sum_{j=1}^N \mathbb{I}[\neg \text{abstain}_{j,\text{remove}}]. \quad (4)$$

Counterfactual abstention shift is $\text{CAS} = P(\text{abstain} \mid \text{necessary removal}) - P(\text{abstain} \mid \text{full})$. A system that tracks necessity should have high PBS, low UPR, positive CAS, and high stability under `irrelevant_removal`.

Secondary metrics. Retrieval is evaluated with Recall@k, nDCG@k, and *Complete Evidence Recall*: whether at least one entire MSES was retrieved. This avoids over-crediting a retriever that retrieves individual relevant passages but misses a necessary multi-passage chain. Generation evaluation includes atomic claim precision/recall, citation precision, citation completeness, selective accuracy, risk–coverage curves, Brier score for answerability probability, latency, tokens, tool calls, and cost.

Table 1: Public IPO Finance Agent question taxonomy at the audited commit. The 70-item public set is a useful seed pool, but its single-issuer concentration motivates multi-filing construction for EVISUFF-FINANCE.

Original category	Questions
Quantitative	28
Disclosure	21
Forensic	8
Comparative	6
Governance	4
Modeling	3
Total	70

Table 2: Descriptive audit of upstream automatic rubric artifacts. Recommendation labels and quality scores are reported exactly as stored upstream; they are not independent human labels.

Artifact	Records	Mean quality	Recommendations
Checked	70	0.763	44 repair / 21 enrich / 5 stop
Seventh pass	70	0.896	43 repair / 4 enrich / 23 stop

Statistical analysis. We will bootstrap at the question level to preserve paired conditions, report 95% confidence intervals, use McNemar tests for matched behavior changes, and fit mixed-effects logistic models with model and question/filing effects where appropriate. Holm correction applies to secondary families of hypotheses. LLM judging is auxiliary until calibrated against at least 150 human-labeled claim–citation pairs, with judge–human agreement and an error analysis reported.

6 Implementation and planned baselines

The accompanying package validates JSONL annotations, computes a minimum hitting set, materializes all four conditions, calculates paired metrics, and provides a question-level bootstrap. Unit tests cover multiple evidence paths, condition construction, retrieval completeness, invalid inputs, and paired metrics. A fictional fixture is included solely to test software behavior.

The planned retrieval controls are BM25, dense BGE-M3, reciprocal-rank-fusion hybrid retrieval, contextual hybrid retrieval with reranking, and a gold evidence oracle. Retrieval parameters (512/1,024-token chunks, 128-token overlap, $k \in \{5, 10, 20\}$) are tuned on development data and then frozen. The generation matrix will include models from at least three frontier providers and one open-weight 8B–14B baseline, with temperature zero and three repetitions. The full study has $60 \times 4 \times 5 \times 3 = 3,600$ generations before optional ablations.

The scientific contribution is not that one retriever dominates another. Retrievers are controlled factors needed to separate an evidence-retrieval failure from a failure to use evidence supplied to the model. The `gold_only` condition is the key oracle for this decomposition.

7 Preliminary reproducibility audit

We audited public artifacts from IPO Finance Agent at commit `091f069`; the full revision is recorded in the released audit JSON. This audit is descriptive: it neither independently validates the benchmark’s rubrics nor estimates model accuracy. It establishes what can and cannot be treated as a gold-evidence source when constructing EVISUFF-FINANCE.

The public repository includes two automatically checked rubric artifacts (Table 2). The later pass reports higher mean overall quality, but also contains 23 `stop` and 43 `repair` recommendations. No record exposes a top-level evidence-identifier field. These upstream fields must not be interpreted as human-validation error rates; they motivate a separate, stable, claim-to-passage annotation layer.

One public Qwen result artifact contains 70 successful outputs. Sixty-eight contain at least one URL (median one URL; maximum ten), but none contains a Markdown inline citation or a stable

evidence/chunk/passage ID token. This is a format-level observation about one artifact, not a claim about factual correctness, all models, or IPO Finance Agent as a whole. It does show why a document-level URL cannot by itself establish citation completeness for an atomic claim.

Finally, eight unit tests pass and the fictional fixture validates without schema errors. Its smoke test distinguishes an intentionally evidence-sensitive policy from an intentionally relevance-only policy. Those values are software verification only; they are not included as benchmark results and must not be used to support an empirical claim.

8 Discussion and limitations

The core risk is the validity of the human evidence sets. Minimality is judgment-dependent, SEC filings can contain redundant disclosures, and parametric model knowledge may permit a correct answer even after a controlled removal. We mitigate rather than eliminate these risks through multiple MSES paths, closed-book controls, frozen corpora, independent verification, and an explicitly bounded claim: behavior is evaluated relative to the allowed corpus.

The financial domain is valuable because filings are long, consequential, and traceable, but it limits generalization. The intervention protocol should later be tested on legal, biomedical, and technical standards documents. Temporal invalidity and contradictions are also important in finance; they are left as post-core extensions rather than being folded into the first study.

This V0.1 is not ready for a NeurIPS submission. It has no completed human-validated dataset and no real-model results. Its purpose is to make the future experiment falsifiable, auditable, and difficult to overclaim. The necessary next milestone is a double-annotated 20-item pilot meeting the pre-specified quality gates.

9 Conclusion

EVISUFF-FINANCE asks a narrower question than generic financial-agent quality: does the agent change from supported answering to abstention when all known necessary evidence paths are broken? The proposed paired design distinguishes relevance from sufficiency, retrieval from evidence use, and citation presence from claim-level support. The released V0.1 implementation and audit provide a reproducible foundation; empirical conclusions await the planned human annotation and model study.

Research transparency statement

The implementation, fictional fixture, protocol, and reproducible upstream audit are public. No private filing data, API keys, held-out questions, or model results are included. The author list, venue, final dataset, and model access details remain to be confirmed. A complete official checklist must be generated for the final selected NeurIPS or workshop track.

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A Annotation record requirements

Each record contains a question, a bounded allowed corpus, a reference answer, atomic claims, evidence units, all known MSES paths, distractors, answerability, temporal metadata, and adjudication metadata. The JSONL schema and validation command are released with the code. Questions with unresolved alternative evidence paths are excluded rather than forced into a necessary-removal intervention.

B Reproducibility checklist for the future empirical study

The final empirical release will provide filing hashes, data and prompt versions, retrieval configuration, exact model IDs and access dates, run IDs, costs, evaluator calibration, scripts for bootstrap intervals and tests, and an error-analysis sample. This appendix is a planning commitment, not evidence that the future experiment has already been performed.